

XIN YAO

1928 SW 42nd Drive, Gainesville, FL 32607, USA
(+1) 571-236-1806 · xyao@ufl.edu · xyao2021.github.io/homepage
Google Scholar · LinkedIn · GitHub

RESEARCH INTERESTS

My research lies at the intersection of machine learning, wireless communication systems, and information theory. I combine theoretical analysis with algorithm and system design to develop efficient, secure, and adaptive methods for distributed learning and dynamic network control, and evaluate them on real-world wireless testbeds that I build from the physical layer up.

Keywords: reinforcement learning for wireless networks; Lyapunov optimization under real-world constraints; LoRa and low-power multi-hop networks; software-defined radio and physical-layer implementation; wireless testbeds and remote experimentation platforms; decentralized and federated learning; error feedback and communication compression; information-theoretic secure aggregation; mmWave and RIS-assisted systems; wireless sensing and indoor localization.

RESEARCH STATEMENT (SUMMARY)

The long-term goal of my research is to make learning-driven wireless networks practical: networks that use machine learning to allocate resources, share spectrum, and adapt the physical layer, with guarantees of convergence, privacy, and reliability that continue to hold after deployment on real hardware. A central motivation is the gap between simulation and reality: many learning algorithms proposed for wireless systems perform well in simulation but degrade on real hardware, where carrier-frequency offsets, timing errors, interference, and hardware outages violate their design assumptions. I address this gap from both directions, by developing algorithms whose analysis accounts for real-world constraints and by building experimental infrastructure that makes over-the-air validation systematic. My doctoral research follows three threads: (1) *adaptive error feedback for distributed learning*, where A-DEFEAT compensates for communication-compression error in decentralized learning with a new proof technique and convergence analysis; (2) *reinforcement learning for real-world wireless networks*, where objectives are derived from Lyapunov functions that encode real-world constraints, state representations are built from quantities a deployed system can observe, and Q-learning, DDPG, and MADDPG policies are compared in simulation before deployment (spectrum sharing on a real-time LoRa testbed, LDPP-MADDPG power allocation, and FORGE-LoRa multi-hop adaptation); and (3) *wireless testbed development*, including the physical-layer stack of the EdgeAI Lab testbed, the UnionLabs remote-access platform, and UAV nodes.

Future plans build on these results in three directions: cross-layer learning-based adaptation for heterogeneous networks with safe online learning under non-stationary conditions; digital-twin pipelines calibrated from testbed measurements, with quantitative criteria for when a policy trained in the twin can be deployed over the air; and decentralized learning over unreliable wireless networks with time-varying topologies, lossy links, and quantized communication, connected to secure aggregation under user dropouts. In parallel, I plan to develop UnionLabs into shared community infrastructure federated across institutions, with support sought from NSF CNS/NeTS and CCF programs, a CAREER proposal, and industry partnerships in AI-for-RAN.

EDUCATION

University of Florida, Gainesville, FL, USA

08/2024 – present

Ph.D. Candidate, Electrical and Computer Engineering

Advisor: Prof. Mingyue Ji (EdgeAI Lab)

Research: machine learning for real-world wireless systems; learning-driven network control on live testbeds;

decentralized learning

University of Utah, Salt Lake City, UT, USA

08/2021 – 07/2024

Ph.D. Studies, Electrical and Computer Engineering (transferred to the University of Florida with my advisor's research group in Aug. 2024)

Advisor: Prof. Mingyue Ji

George Washington University, Washington, DC, USA

01/2020 – 06/2021

M.S., Electrical Engineering

Tianjin University of Science and Technology (TUST), Tianjin, China

09/2015 – 06/2019

B.Eng., Electronic Information Engineering

RESEARCH EXPERIENCE

Doctoral Researcher, EdgeAI Lab (Advisor: Prof. Mingyue Ji)

08/2021 – present

University of Utah (08/2021 – 07/2024) and University of Florida (08/2024 – present)

EdgeAI Lab: Real-World Wireless Communication Testbed

University of Florida · Platform for the over-the-air experiments in [C1], [C4] · Submitted manuscript [S2]

- Implemented the physical-layer software stack of the lab's wireless testbed in C++: modulation and demodulation, filtering, synchronization, channel encoding and decoding, and correction of carrier-frequency offset and phase offset.
- Built Python APIs on top of the C++ core so that over-the-air experiments can be scripted, repeated, and parameterized from a single configuration; designed automated experiment workflows that adapt to heterogeneous physical-layer hardware (USRP software-defined radios and LoRa radios).
- Deployed the group's learning algorithms (reinforcement-learning spectrum sharing, power allocation, and per-slot LoRa PHY adaptation) on both lab-scale and building-scale deployments of the testbed, and used the platform for the real-time LoRa experiments reported in [C4] and the multi-hop LoRa experiments in [C1].
- Developed SONAR, a hardware-in-the-loop approach to synthesizing wireless systems [S2].
- Technologies: C++, Python, USRP software-defined radios, LoRa radios, Linux.

UnionLabs: AWS-Based Remote Access and Sharing of NextG and IoT Testbeds

University of Florida · NSF-funded project (CIRC award, 2025–2027) · Poster [P2], demo [P3]

- Developed and deployed the AWS-hosted web user interface for remote access to the wireless testbeds.
- Designed the physical-layer (PHY) code for different hardware, including USRP software-defined radios and LoRa devices.
- Designed the unified *Union API* that reads and drives the different PHY implementations and connects them automatically to users' learning or optimization algorithms, and an abstraction layer that lets users run the PHY and Union API with a single command.
- Supported collaborators from several universities in running their learning and optimization algorithms through UnionLabs, including experiments that use multiple testbeds at the same time.
- The UnionLabs and OSWireless demonstration won *Third Prize* in the Demo Competition of the Warren B. Nelms Annual IoT Conference (Dec. 2025).
- Technologies: AWS, Java, JavaScript, JSON, Python, USRP, LoRa.

Constrained Optimization for Real-World Wireless Problems with Reinforcement Learning

University of Utah and University of Florida · Publications [C4], [C5]; poster [P4]; submitted manuscript [S4]

- Derived optimization objectives with Lyapunov drift-plus-penalty (LDPP) functions under real-world constraints, turning long-term constrained problems into per-slot decisions that a learning agent can act on.
- Designed novel state representations for reinforcement-learning algorithms applied to communication-system control, and evaluated Q-learning, DDPG, and MADDPG policies both in simulation and in deployment on the EdgeAI Lab testbed.
- Proposed LDPP-MADDPG, a distributed power-allocation scheme for mmWave cellular networks that combines a Lyapunov drift-plus-penalty objective with multi-agent DDPG [C5]; an earlier MADDPG power-allocation study used measurement data from the POWDER platform [P4].
- Carried out an experimental evaluation of reinforcement learning for spectrum sharing on a real-time LoRa testbed, comparing Q-learning, DDPG, and MADDPG on live hardware [C4].
- Developed LYRIC, a Lyapunov-constrained multi-agent power-control method evaluated on measured LoRa data [S4].
- Technologies: Python, reinforcement learning, Lyapunov optimization.

Resilient Multi-Hop LoRa Networks: FORGE-LoRa, HYDRA-LoRa, and CONCERT-LoRa

University of Florida · Publication [C1] (IEEE MILCOM 2026, accepted); submitted manuscripts [S1], [S3]

- Derived the optimization objective for multi-hop LoRa networks with Lyapunov functions under real-world constraints.
- Designed the system architecture for jointly adapting the LoRa PHY parameters on a per-slot basis.
- Building on the joint-adaptation method, designed a path discovery and optimization system that keeps multi-hop communication always live against interference and hardware outages.
- Evaluated the dynamic-control policies in simulation and in deployment on the EdgeAI Lab testbed across multiple reinforcement-learning algorithms.
- Extended the LoRa mesh work to anti-jamming operation: HYDRA-LoRa, anti-jamming multi-path selection for LoRa mesh networks [S1], and CONCERT-LoRa, certified concurrency for anti-jamming LoRa mesh networks [S3].
- Technologies: LoRa, Python, reinforcement learning, Lyapunov optimization.

LoFiLoc: Device-Free LoRa Fingerprint Indoor Localization with Commodity Radios

University of Florida · Publication [C2] (IEEE MILCOM 2026 Workshops, accepted)

- Device-free indoor localization based on fingerprints of LoRa channel measurements collected with commodity LoRa radios, so that a target carrying no device can be localized from changes in the wireless channel.

Transformer-Enhanced DDPG for RIS-Assisted MU-MISO Systems (collaboration)

University of Florida · Publication [C3] (IEEE MILCOM 2026 Workshops, accepted)

- Contributed to a transformer-enhanced deep deterministic policy gradient (DDPG) method for the joint design of base-station beamforming and reconfigurable-intelligent-surface (RIS) phase shifts in multi-user MISO systems, with H. Zhang and O. Jiang.

A-DEFEAT: Adaptive Error Feedback for Decentralized Learning

University of Utah and University of Florida · Poster [P1]; manuscript [M1]

- Designed a new decentralized learning algorithm with an adaptive, discounted error-feedback (error-compensation) mechanism for communication compression under data heterogeneity.
- Proposed the key proof technique for the adaptive error-feedback mechanism, established convergence guarantees for the proposed algorithm, and compared its convergence rate with existing algorithms.
- Evaluated the algorithm against existing decentralized learning methods under a range of settings.
- Technologies: Python, optimization theory.

Information-Theoretic Secure Aggregation with Uncoded Groupwise Keys

University of Utah · Publications [J1], [C6] · With K. Wan, H. Sun, M. Ji, and G. Caire

- Studied the information-theoretic secure aggregation problem of federated learning, in which the server must learn only the aggregate of the users' models while tolerating user dropouts, under the practical assumption that keys are mutually independent and each key is shared, uncoded, by a group of S users.
- The work characterizes when uncoded groupwise keys are optimal: if $S > K - U$ (with K users and at most $K - U$ dropouts) a new scheme achieves the same optimal communication cost as the best coded-key scheme, whereas if $S \leq K - U$ uncoded groupwise key sharing is strictly sub-optimal.
- Implemented the proposed secure-aggregation scheme on Amazon EC2 and compared it experimentally with existing secure-aggregation schemes that use offline key sharing.

UAV-Based Wireless Communication Platform

University of Florida

- Assembled customized drones and integrated wireless communication payloads for airborne networking experiments; initialized and tuned the flight controllers; designed and mounted the customized communication payloads.
- Designed the communication protocol between ground devices and the airborne node.
- Technologies: UAV flight controllers, embedded systems, wireless payload integration.

Master's Projects

George Washington University

- **Age of Information with Energy Harvesting** (advisor: Prof. Omur Ozel). Analyzed the age of information (AoI) in M/G/1/1 and idealized infinite-buffer queueing models, including variants with packet deadlines, and investigated AoI-minimization strategies for the bufferless M/G/1/1 model with packet deadlines.
- **Secure File Transfer: Encryption and Decryption for FTP** (advisor: Prof. Tian Lan). Designed a custom encryption/decryption scheme for FTP on Linux with client-side encryption and server-side decryption of commands and data, and verified end-to-end encrypted FTP transmission.

Undergraduate Projects

Tianjin University of Science and Technology

- **Voice-controlled smart-home system** (advisor: Jing He). Designed STM32F103-based hardware with an LD3320 voice-recognition module; wrote firmware linking the microcontroller to speaker, PC, and LCD peripherals; tested spoken commands with system feedback.
- **License-plate recognition system** (advisor: Prof. Yang An). Built a MATLAB pipeline for image capture from surveillance cameras, filtering (Hamming window), and character segmentation and matching.
- **Android music player** (advisor: Prof. Ya-Jun Li). Developed a multi-format music player in Java using Eclipse and the Android SDK.
- **Digital counter** (advisor: Prof. Yu-liang Liu). Designed the counter circuit, selected components, and assembled, tested, and debugged the board.

PUBLICATIONS

Google Scholar: scholar.google.com/citations?user=gwTaRUkAAAAJ. My name is shown in bold.

Conference Papers

- [C1] **X. Yao**, A. Bhuyan, and M. Ji, "FORGE-LoRa: Joint per-Slot PHY Adaptation for Multi-Hop LoRa Networks," in *Proc. IEEE Military Communications Conference (MILCOM)*, National Capital Region, USA, Oct. 2026.

(Accepted, to appear.)

Joint per-slot adaptation of LoRa PHY parameters combined with path discovery and optimization so that multi-hop LoRa communication stays live under interference and hardware outages; evaluated in simulation and on the EdgeAI Lab testbed.

- [C2] **X. Yao**, A. Bhuyan, and M. Ji, “LoFiLoc: Device-Free LoRa Fingerprint Indoor Localization with Commodity Radios,” in *Proc. IEEE Military Communications Conference (MILCOM) Workshops*, National Capital Region, USA, Oct. 2026. (Accepted, to appear.)
- [C3] H. Zhang, O. Jiang, **X. Yao**, and M. Ji, “Transformer-Enhanced DDPG for Beamforming and Phase Shift in RIS-Assisted MU-MISO Systems,” in *Proc. IEEE Military Communications Conference (MILCOM) Workshops*, National Capital Region, USA, Oct. 2026. (Accepted, to appear.)
- [C4] **X. Yao**, X. Wang, H. Zhang, and M. Ji, “Experimental Evaluation of Reinforcement Learning for Spectrum Sharing on a Real-Time LoRa Testbed,” in *Proc. IEEE International Symposium on Spectrum Innovation (DySPAN)*, Washington, DC, USA, May 2026, pp. 1–8. DOI: 10.1109/DySPAN69846.2026.11571139.
Q-learning, DDPG, and MADDPG spectrum-sharing policies evaluated on live LoRa hardware in real time rather than in simulation only.
- [C5] **X. Yao**, A. Bhuyan, X. Zhang, and M. Ji, “A Novel LDPP-MADDPG Approach for Distributed Power Allocation in mmWave Cellular Networks,” in *Proc. IEEE Military Communications Conference (MILCOM), 6G and Spectrum Security for Critical Communication (6GSECC) Workshop*, Los Angeles, CA, USA, Oct. 2025.
Distributed power allocation for mmWave cellular networks that combines a Lyapunov drift-plus-penalty (LDPP) objective with multi-agent deep deterministic policy gradient (MADDPG).
- [C6] K. Wan, **X. Yao**, H. Sun, M. Ji, and G. Caire, “GroupSecAgg: Information Theoretic Secure Aggregation with Uncoded Groupwise Keys,” in *Proc. IEEE International Conference on Communications (ICC)*, Rome, Italy, May–Jun. 2023, pp. 3890–3895. DOI: 10.1109/ICC45041.2023.10279171.
Conference version of [J1].

Journal Articles

- [J1] K. Wan, **X. Yao**, H. Sun, M. Ji, and G. Caire, “On the Information Theoretic Secure Aggregation with Uncoded Groupwise Keys,” *IEEE Transactions on Information Theory*, vol. 70, no. 9, pp. 6596–6619, Sept. 2024. DOI: 10.1109/TIT.2024.3422087. Preprint: arXiv:2204.11364.
Characterizes the optimal communication cost of information-theoretic secure aggregation when keys are mutually independent and each key is shared by a group of S users: uncoded groupwise keys are optimal when $S > K - U$ and strictly sub-optimal when $S \leq K - U$. Includes an Amazon EC2 implementation compared against existing schemes with offline key sharing.

Submitted Manuscripts (Under Review)

- [S1] **X. Yao**, A. Bhuyan, and M. Ji, “HYDRA-LoRa: Anti-Jamming Multi-Path Selection for LoRa Mesh Networks,” submitted to *IEEE International Conference on Communications (ICC)*, 2027.
- [S2] **X. Yao** and M. Ji, “SONAR: Hardware-in-the-Loop Synthesis of Wireless Systems,” submitted to *IEEE International Conference on Communications (ICC)*, 2027.
- [S3] **X. Yao**, A. Bhuyan, and M. Ji, “CONCERT-LoRa: Certified Concurrency for Anti-Jamming LoRa Mesh Networks,” submitted to *IEEE Internet of Things Journal*.
- [S4] **X. Yao**, X. Zhang, A. Bhuyan, and M. Ji, “LYRIC: Lyapunov-Constrained Multi-Agent Power Control with Measured LoRa Evaluation,” submitted to *IEEE Transactions on Cognitive Communications and Networking*.

Other Manuscripts

- [M1] **X. Yao**, S. Wang, J. Perazzone, K. S. Chan, and M. Ji, “Compressed Decentralized Learning with Error-Feedback under Data Heterogeneity,” 2026.

PRESENTATIONS, POSTERS AND DEMONSTRATIONS

Conference Presentations

- [T1] “FORGE-LoRa: Joint per-Slot PHY Adaptation for Multi-Hop LoRa Networks,” IEEE Military Communications Conference (MILCOM), National Capital Region, USA, Oct. 2026. (Upcoming.)
- [T2] “Experimental Evaluation of Reinforcement Learning for Spectrum Sharing on a Real-Time LoRa Testbed,” IEEE International Symposium on Spectrum Innovation (DySPAN), Washington, DC, USA, May 2026.
- [T3] “A Novel LDPP-MADDPG Approach for Distributed Power Allocation in mmWave Cellular Networks,” IEEE MILCOM 6G and Spectrum Security for Critical Communication (6GSECC) Workshop, Los Angeles, CA, USA, Oct. 2025.
- [T4] “GroupSecAgg: Information Theoretic Secure Aggregation with Uncoded Groupwise Keys,” IEEE International Conference on Communications (ICC), Rome, Italy, Jun. 2023.
- [T5] “High-Band Frequency Spectrum for 5G” (team presentation), George Washington University, Washington, DC, USA, 2020.

Posters and Demonstrations

- [P1] “A-DEFEAT: An Adaptively Discounted Error-Feedback Algorithm for Decentralized Learning” (poster), North American School of Information Theory (NASIT), Brigham Young University, Provo, UT, USA, Jun. 2026.
- [P2] “UnionLabs: Shared Access to Heterogeneous Wireless IoT and Wireless OS Testbeds” (poster), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2025.
- [P3] “OSWireless: Automatic System to Maximize Network Throughputs” (live demonstration), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2025. *Third Prize, Demo Competition.*
- [P4] “MADDPG-Based Power Allocation Scheme Using Measurement Data from POWDER” (poster), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2024.

TEACHING EXPERIENCE

- **Lab Instructor, EEL 4514 Communication Systems and Components, University of Florida.** Designed and verified the lab sessions on signal processing, held and graded the lab sessions, and held office hours for course-related questions before exams.
- **Teaching Assistant, ECE 5510 Random Processes, University of Utah.** Assisted in designing the lecture slides, graded homework and exams, and held weekly office hours to answer course-related questions.
- **Project Instructor, UnionLabs Web Design (undergraduate project), University of Florida.** Directed and guided students through the web-design part of the UnionLabs project, from background knowledge to design skills including JavaScript, ReactJS, and the communication logic between the web interface and the testbeds.
- **Project Instructor, Automatic System Design (undergraduate project), University of Florida.** Guided students to understand the structure and code of the telco system, an AI-in-the-loop evaluation system, and how to use the code for customized purposes, and introduced hardware-in-the-loop design based on the telco system.
- **Testbed Training and Onboarding, UnionLabs.** Onboard collaborators from multiple universities, ranging from machine-learning researchers new to radios to communications engineers new to reinforcement learning, onto the shared wireless testbeds: prepare documentation for the customized system, design worked examples, and structure first sessions so that every user runs a real over-the-air experiment quickly.

TEACHING STATEMENT (SUMMARY)

Philosophy. Students learn engineering most effectively when theory is connected to the physical world as early and as often as possible. In wireless communications and machine learning this connection is concrete: a matched filter derived on paper can be observed as a constellation diagram from the student's own transmitter, a coding theorem can be verified as a measured shift in a bit-error-rate curve, and a learning algorithm can be run and measured on real machines. My goal is to help students build a solid conceptual foundation, the practical skill to implement and debug real systems, and the confidence that comes from seeing their own analysis confirmed by measurement.

Methods. Theoretical foundations in class, with emphasis on the mathematical expressions; live software-defined-radio demonstrations so that effects such as intersymbol interference and coding gain are observed as they are derived; project-based homework tested over the air or against recorded real signals through remote testbed access of the kind UnionLabs provides; and projects scaffolded in explicit stages from simulation to hardware, so that students with different preparation all reach the final over-the-air milestone. Learning is assessed through staged milestones that reward implementation, measurement, and debugging alongside analysis, and the course is adjusted using student feedback throughout the semester.

Mentoring. Early contact with the real system followed by growing ownership: a new student runs a working over-the-air experiment in the first weeks, then takes responsibility for one testbed component or algorithm, growing from reproducing a known result to extending it to leading a first-authored paper, with regular individual meetings, projects matched to each student's strengths, and frequent presentations.

Courses I am prepared to teach. Signals and systems, digital communications, and digital signal processing (undergraduate); wireless system design and networking courses that connect the physical and MAC layers; machine learning for wireless networks (graduate); and a new project-based course, *Hands-On Wireless Systems with Software-Defined Radios*, in which students implement a complete communication link and then apply a learning algorithm to control it.

PROFESSIONAL SERVICE

- Judge, Gator Quant Hack Competition 2026
- Student Volunteer, IEEE Military Communications Conference (MILCOM), National Capital Region, USA 2026
- Student Volunteer, IEEE International Conference on Computer Communications (INFOCOM) 2022

TECHNICAL SKILLS

- **Programming:** Python, C/C++, Java, JavaScript, MATLAB, Arduino, JSON, XHTML/DHTML.
- **Machine learning:** reinforcement learning (Q-learning, DDPG, MADDPG, transformer-enhanced policies), federated learning, decentralized learning, communication compression and error feedback, secure aggregation, distributed and parallel computing.
- **Wireless communications and signal processing:** physical-layer implementation (modulation and demodulation, synchronization, filtering, channel encoding and decoding, carrier-frequency and phase-offset correction), software-defined radio (USRP), LoRa and low-power multi-hop networking, mmWave power allocation, RIS-assisted systems, real-world testbed development, network protocol analysis, wireless fingerprinting for localization.
- **Optimization and theory:** Lyapunov drift-plus-penalty optimization, convergence analysis of decentralized learning, information-theoretic secure aggregation, queueing models and age of information.
- **Embedded, cloud, and systems:** STM32 microcontrollers, Android SDK (Eclipse/JDK), AWS (EC2 and web deployment), Linux, UAV flight-controller setup and payload integration.
- **Design and productivity tools:** AutoCAD, Siemens NX, Sharp3D, Adobe (Photoshop, Premiere Pro, After Effects, Acrobat), MS Office, L^AT_EX.

HONORS AND AWARDS

- Overall Runner-Up (Second Prize), NSF SpectrumX Student Data & Algorithm Competition (SpX-DAC), team with X. Wang and Y. Hu, advised by Prof. Mingyue Ji; 37 teams from 25 institutions 2026
- Third Prize, Demo Competition, Warren B. Nelms Annual IoT Conference, University of Florida (UnionLabs / OSWireless demonstration) 2025
- Third-Prize Scholarship, Tianjin University of Science and Technology 2017
- Advanced Individual of Social Work, Tianjin University of Science and Technology 2016
- Excellent League Member, Tianjin University of Science and Technology 2016

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